

EMCH 260

SOLID MECHANICS

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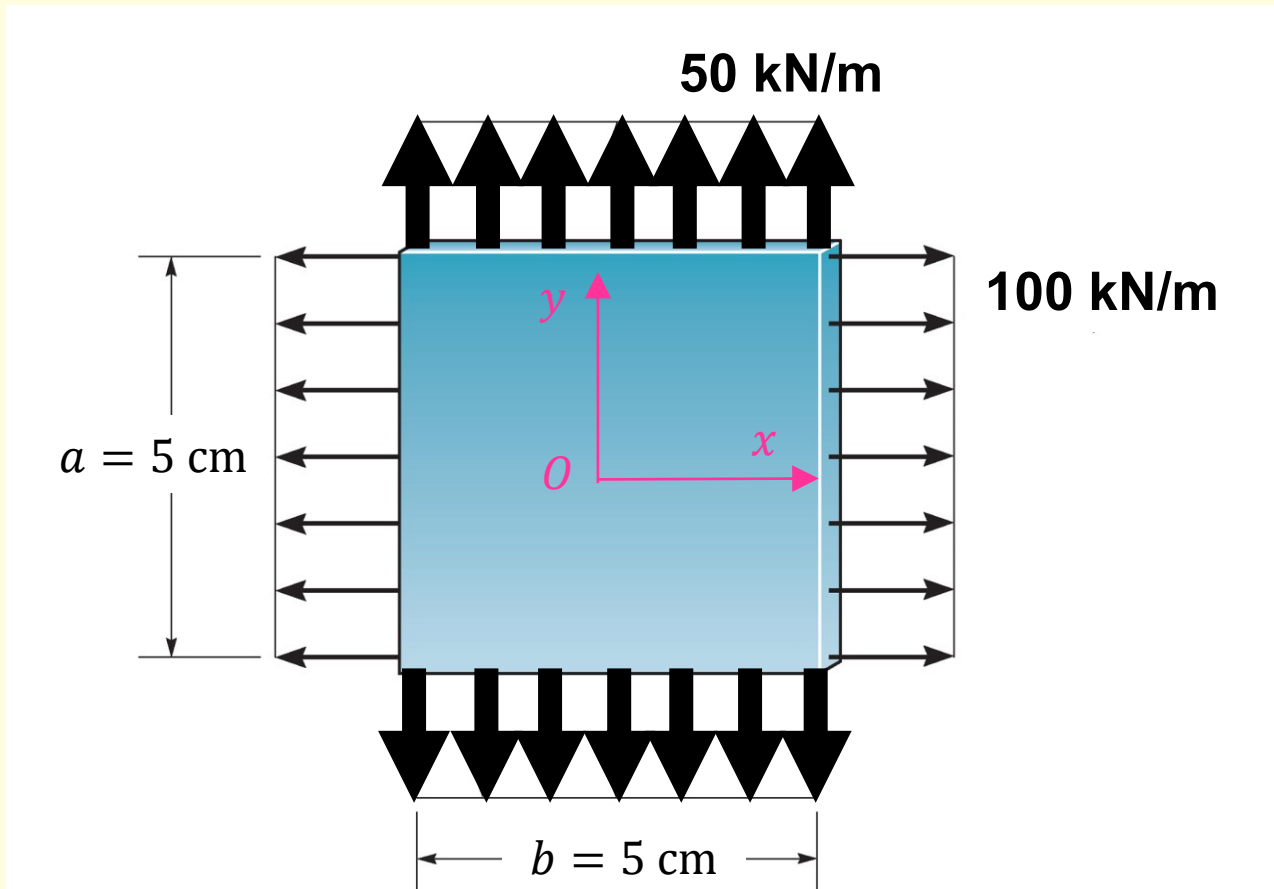
Strain Transformation

Chapter 10.4

- ✓ Plane stress and plane strain for combined loadings and complex deformation problems

IN-CLASS EXAMPLE 1

A square plate in biaxial stress is subjected to tensile force 100 kN/m and 50 kN/m in x- and y-direction, respectively. The plate has $E = 20$ GPa and Poisson's ratio $\nu = 0.25$. If the plate is originally square and has dimensions of $a = b = 5$ cm, and a thickness of $t = 0.5$ cm, determine the change in the volume of the plate (ΔV) and the **absolute maximum shear strain** ($\gamma_{\max}^{\text{abs}}$) in the plate.



IN-CLASS EXAMPLE 1, cont'd

Solutions:

The two normal stresses in the plate are:

$$\sigma_x = \frac{10^5 \times 0.05}{0.05 \times 0.5 \times 10^{-2}} = 20 \text{ MPa}$$
$$\sigma_y = \frac{50 \times 10^3 \times 0.05}{0.05 \times 0.5 \times 10^{-2}} = 10 \text{ MPa}$$

This is a plane stress problem, NOT plane strain!

Since there is no shear stress when using this Oxy system, the above stresses are the two principal stresses and then the Mohr's circle for the plane stress is constructed!

$$\sigma_1 = \sigma_x = 20 \text{ MPa}$$

$$\sigma_2 = \sigma_y = 10 \text{ MPa}$$

IN-CLASS EXAMPLE 1, cont'd

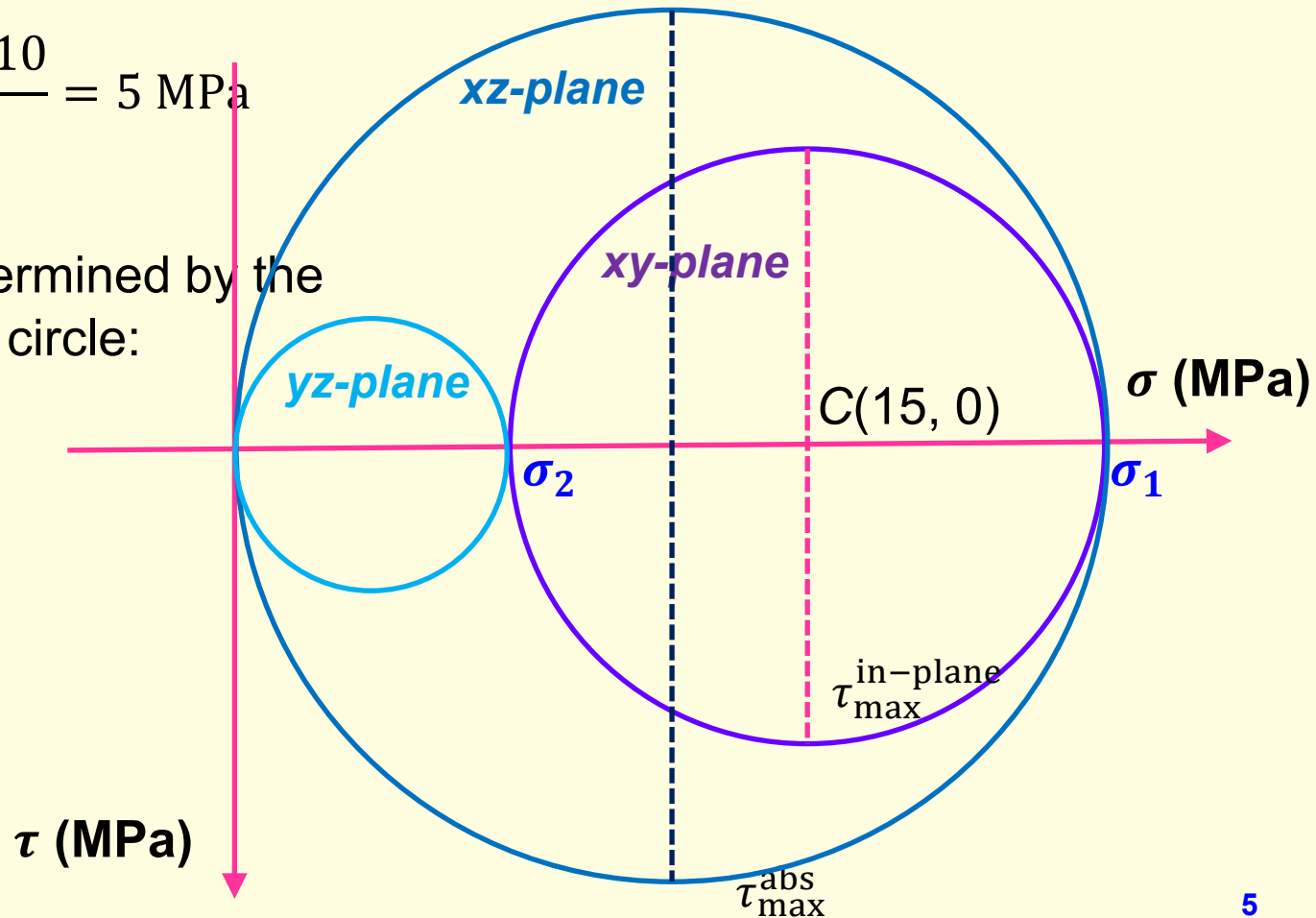
Solutions, cont'd:

The radius of the Mohr's circle is:

$$R = \frac{\sigma_1 - \sigma_2}{2} = \frac{20 - 10}{2} = 5 \text{ MPa}$$

$\tau_{\max}^{\text{in-plane}}$ can be determined by the radius of the Mohr's circle:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = 10 \text{ MPa}$$



IN-CLASS EXAMPLE 1, cont'd

Solutions, cont'd:

According to the Hooke's law, to get the maximum in-plane shear strain, we need to know the shear modulus, which can be obtained by the material property relationship:

$$G = \frac{E}{2(1 + \nu)} = \frac{20 \times 10^9}{2 \times (1 + 0.25)} = 8 \times 10^9 \text{ Pa}$$

$$\rightarrow \gamma_{\max}^{\text{abs}} = \frac{\tau_{\max}^{\text{abs}}}{G} = \frac{10 \times 10^6}{8 \times 10^9} = 1.25 \times 10^{-3} \text{ rad.} \quad \textbf{(Ans.)}$$

Note that the maximum shear strain occurs in **xz-plane**.

IN-CLASS EXAMPLE 1, cont'd

Solutions, cont'd:

According to the generalized Hooke's law, the strain in three directions are:

$$\varepsilon_x = \frac{1}{E} [\sigma_x - \nu \cdot (\sigma_y + \sigma_z)] = \frac{20 \times 10^6 - 0.25 \times (10 \times 10^6)}{20 \times 10^9} = 0.875 \times 10^{-3}$$

$$\varepsilon_y = \frac{1}{E} [\sigma_y - \nu \cdot (\sigma_x + \sigma_z)] = \frac{10 \times 10^6 - 0.25 \times 20 \times 10^6}{20 \times 10^9} = 0.25 \times 10^{-3}$$

$$\varepsilon_z = \frac{1}{E} [\sigma_z - \nu \cdot (\sigma_x + \sigma_y)] = \frac{-0.25 \times (20 + 10) \times 10^6}{20 \times 10^9} = -0.375 \times 10^{-3}$$

Then, the new dimension of the plate is:

$$a' = a \cdot (1 + \varepsilon_y) = 5.00125 \text{ cm}$$

$$b' = b \cdot (1 + \varepsilon_x) = 5.004375 \text{ cm}$$

$$t' = t \cdot (1 + \varepsilon_z) = 0.4998125 \text{ cm}$$

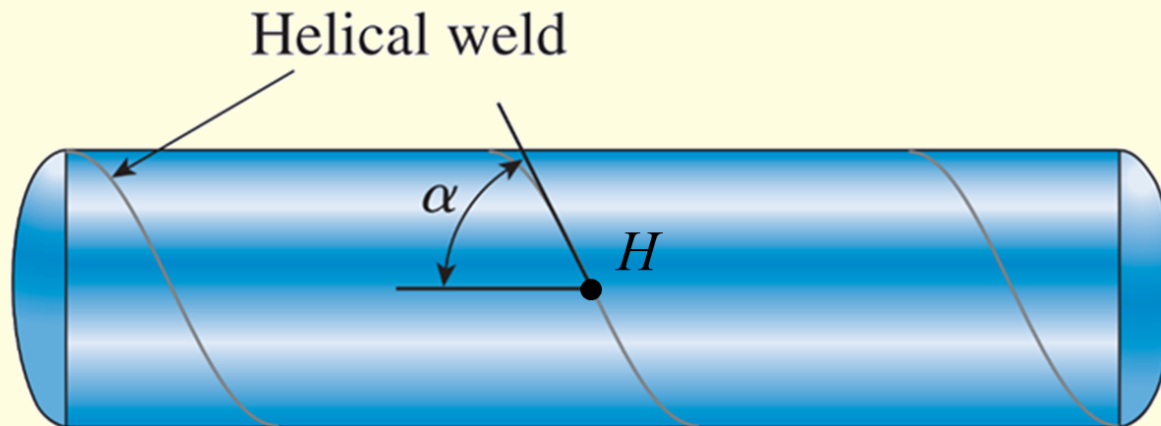
The volume change is:

$$\Delta V = a' \cdot b' \cdot t' - a \cdot b \cdot t = 0.00937 \text{ cm}^3 \quad \textbf{(Ans.)}$$

IN-CLASS EXAMPLE 2

A pressurized steel tank is constructed with a helical weld that makes an angle $\alpha = 55^\circ$ with the longitudinal axis (see figure). The tank has inner radius $r = 0.6$ m, wall thickness $t = 18$ mm, and internal pressure $p = 2.8$ MPa. The steel has modulus of elasticity $E = 200$ GPa and Poisson's ratio $\nu = 0.30$. For a point H on the tank surface, determine the following quantities:

- (a) The circumferential (hoop) and longitudinal stresses.
- (b) The maximum in-plane shear stress and absolute maximum shear stress.
- (c) The normal and shear stresses acting on planes parallel and perpendicular to the weld; show these stresses on a properly oriented stress element.
- (d) The circumferential (hoop) and longitudinal strains.
- (e) The absolute maximum shear strain.



IN-CLASS EXAMPLE 2, cont'd

Solutions

Establish xy -system as shown (z -direction is perpendicular to the surface).

(a) The circumferential (hoop) stress is

$$\sigma_{\text{hoop}} = \frac{pr}{t} = \frac{2.8 \times 10^6 \times 0.6}{0.018} = \boxed{93.333 \text{ MPa}}$$

The longitudinal stress is

$$\sigma_{\text{long}} = \frac{pr}{2t} = \frac{2.8 \times 10^6 \times 0.6}{2 \times 0.018} = \boxed{46.667 \text{ MPa}}$$

(b) The stress in the hoop and longitudinal direction is perpendicular with each other, and there is no shear stress on the point H . Therefore, the two stresses are actually the two principal stresses, i.e.

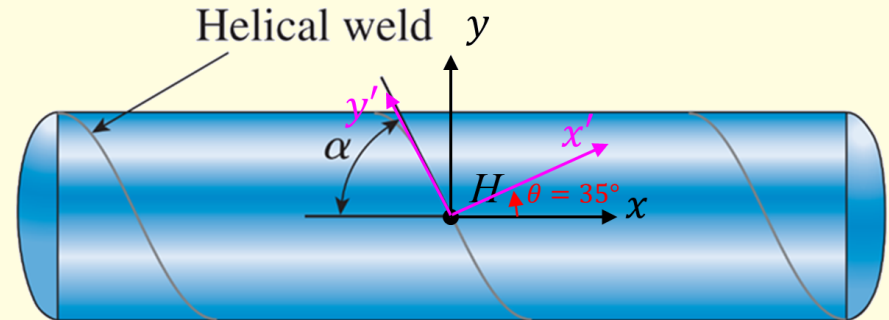
$$\sigma_1 = \sigma_{\text{hoop}} = 93.333 \text{ MPa}$$

$$\sigma_2 = \sigma_{\text{long}} = 46.667 \text{ MPa}$$

Based on the principal stresses, we can draw the Mohr's circle as shown in the figure.

Then, the maximum in-plane stress is

$$\tau_{\text{max}}^{\text{in-plane}} = R = \frac{\sigma_1 - \sigma_2}{2} = \boxed{23.333 \text{ MPa}}$$



IN-CLASS EXAMPLE 2, cont'd

Solutions, cont'd

The absolute maximum stress is

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \boxed{46.667 \text{ MPa}}$$

(c) From the reference point $A(\sigma_x, \tau_{xy}) = A(\sigma_{\text{long}}, 0)$ to the plane along which we want to know the stresses, we need to rotate the CA in the Mohr's circle by $2\theta = 2 \times (90^\circ - \alpha) = 2 \times 35^\circ = 70^\circ$ (counterclockwise).

Using the Mohr's circle, the new stresses correspond to the point P (see figure) are

$$\sigma_{x'} = \sigma_P = \sigma_C - R \cos 2\theta = \frac{46.667 + 93.333}{2} - 23.333 \times \cos 70^\circ = \boxed{62.02 \text{ MPa}}$$

(Tensile)

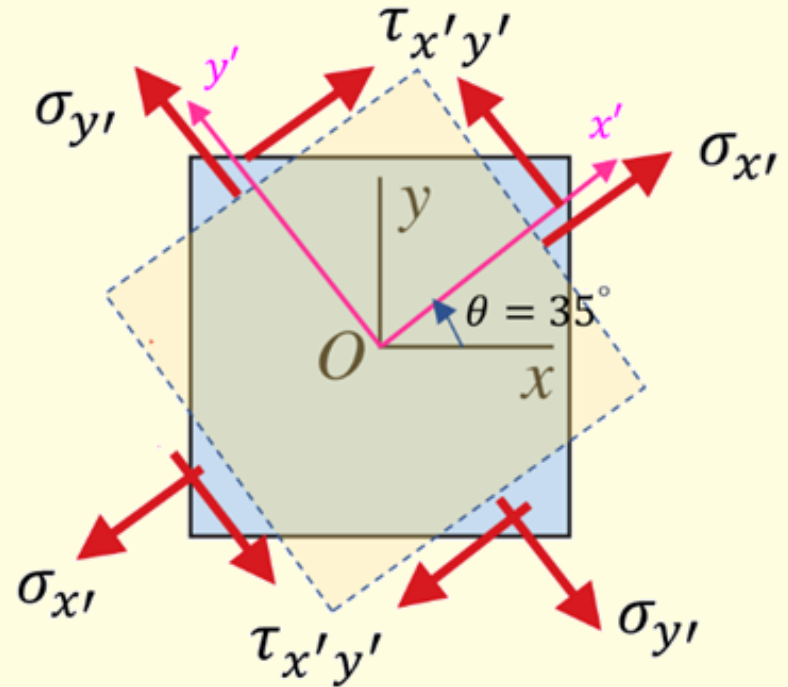
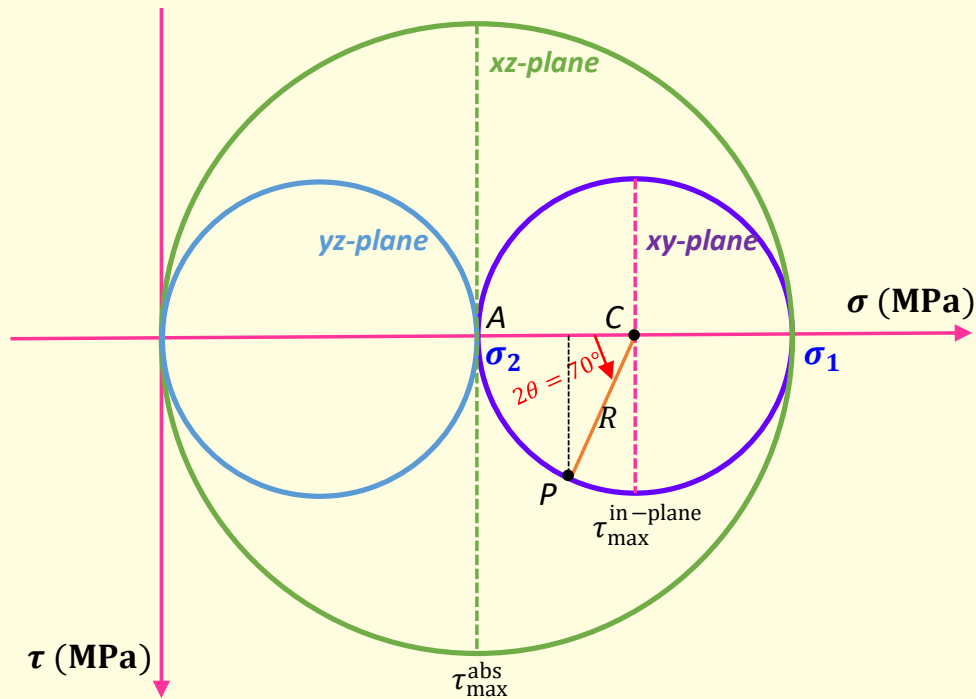
$$\tau_{x'y'} = \tau_P = R \sin 2\theta = 23.333 \times \sin 70^\circ = \boxed{21.926 \text{ MPa}}$$

$$\sigma_{y'} = \sigma_{P'} = \sigma_C + R \cos 2\theta = \boxed{77.980 \text{ MPa}} \text{ (Tensile)}$$

The oriented stress element is shown by the dashed square with stresses labeled.

IN-CLASS EXAMPLE 2, cont'd

Solutions, cont'd



IN-CLASS EXAMPLE 2, cont'd

Solutions, cont'd

(d) For the xyz -coordinate system established, using the generalized 3D Hooke's law, the circumferential (hoop) strain (in y -direction) is

$$\varepsilon_{\text{hoop}} = \frac{\sigma_{\text{hoop}} - \nu\sigma_{\text{long}}}{E} = \frac{93.333 \times 10^6 - 0.3 \times 46.667 \times 10^6}{200 \times 10^9} = \boxed{3.967 \times 10^{-4}}$$

The longitudinal strain (in x -direction) is

$$\varepsilon_{\text{long}} = \frac{\sigma_{\text{long}} - \nu\sigma_{\text{hoop}}}{E} = \frac{46.667 \times 10^6 - 0.3 \times 93.333 \times 10^6}{200 \times 10^9} = \boxed{9.333 \times 10^{-5}}$$

(e) The shear modulus can be obtained by the material property relationship

$$G = \frac{E}{2(1 + \nu)} = \frac{200 \times 10^9}{2 \times (1 + 0.3)} = 76.923 \text{ GPa}$$

Finally, the absolute maximum shear strain is obtained by using the maximum shear stress and Hooke's law:

$$\gamma_{\text{max}}^{\text{abs}} = \frac{\tau_{\text{max}}^{\text{abs}}}{G} = \frac{46.667 \times 10^6}{76.923 \times 10^9} = \boxed{6.07 \times 10^{-4} \text{ rad}}$$

RELEVANT PRACTICE

- **Assignment 4 on Blackboard, problem #5-6 (go to Blackboard under the “Assignments” folder, **Assignment 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**
- **Optional Extra Exercise 4 on Blackboard, problem #4 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 4 due: 8:30 AM, Wednesday, Nov. 17th 2021**)**